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Tsunami evolution and run-up in a large scale experimental facility

V. Sriram [a,*], I. Didenkulova [b,c], A. Sergeeva [c,d], S. Schimmels [e]

[a] Department of Ocean Engineering, IIT Madras, India
[b] Marine Systems Institute, Tallinn University of Technology, Akadeemia tee 15A, 12618 Tallinn, Estonia
[c] Nizhny Novgorod State Technical University n.a. R.E. Alekseev, Minin str. 24, 603950 Nizhny Novgorod, Russia
[d] Institute of Applied Physics, Ulyanov str. 46, 603950 Nizhny Novgorod, Russia
[e] Forschungszentrum Küste (FZK), Merkurstraße 11, 30419 Hannover, Germany

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ABSTRACT

In this paper, we study the propagation and run-up of long tsunami-like waves in the 300 m long Large Wave Flume (GWK), Hannover, Germany and analyze the feasibility of experiments on tsunami run-up in large facilities. This paper is the continuation of our previous paper (Schimmels et al., 2015, companion paper). The propagation of long period waves over large distances for different shapes is studied experimentally and numerically. Fully nonlinear potential flow theory has been used to model these cases numerically along with Korteweg–de Vries simulations. The theoretical explanation of the observed effects has been discussed. The run-up characteristics of the studied waves with respect to their shape and beach slope were also investigated. Further, a down-scaled real tsunami time series is also reproduced experimentally and studied with regard to its possible run-up.

1. Introduction

For modeling of tsunami wave propagation, shoaling and run-up, often solitary or cnoidal waves have been used (Goring, 1979; Synolakis, 1987). However, solitary and cnoidal waves are rarely observed during tsunami propagation. Furthermore, their characteristics are not comparable with most real-field tsunamis as reported by Madsen et al. (2008). Nevertheless, a set of solitary and cnoidal-like waves can be formed in the far field as a result of long wave transformation into an undular bore, as it was observed during the 2004 tsunami in the Indian Ocean near the coast of Thailand (Fig. 1).

More importantly, one cannot generalize a particular tsunami case. In reality, there is a variety of tsunami waves in terms of both periods and wave shapes. Due to the large-distance propagation and complicated bottom and coastal topography, the initial wave changes its shape. Hence, very often even the same tsunami event has very different manifestations in different locations. Therefore, when modeling tsunami, one should also study different wave shapes.

Fig. 2 illustrates the variations in the tsunami profile that occurred along the Japanese coast during the Japan Sea tsunami in 1983. Based on the past tsunami events, tsunamis approaching the shore may broadly be classified as (e.g. Shuto, 1985)

1. non-breaking waves that act as a rapidly rising tide, observed during small and moderate tsunami events after short distance propagation;
2. breaking bore or hydraulic jump (wall of water), observed as a result of wave breaking during large tsunami events after short distance propagation;
3. undular bore, observed after long distance propagation (in terms of wavelength), i.e. the disintegration into series of solitons, see Fig. 1.

The first rigorous solution to the nonlinear shallow water equations was found by Carrier and Greenspan (1958) for non-breaking wave run-up on a plane beach. Based on this approach various shapes of the periodic incident wave trains such as sine wave (Madsen and Fuhrman, 2008), cnoidal wave (Synolakis et al., 1988) and nonlinear deformed periodic wave (Didenkulova et al., 2007a) have been analyzed in the literature. Relevant analysis has also been performed for a variety of solitary waves and single pulses such as soliton (Kânoğlu, 2004; Pedersen and Gjevik, 1983; Synolakis, 1987), sine pulse (Mazova et al., 1991), Lorentz pulse (Pelinovsky and Mazova, 1992), Gaussian pulse (Carrier et al., 2003; Kânoğlu and Synolakis, 2006), *N*-waves (Tadepalli and Synolakis, 1994), “characterized tsunami waves” (Tinti and Tonini, 2005) and a random set of solitons (Brocchini and Gentile, 2001).

Didenkulova and Pelinovsky (2008) and Didenkulova et al. (2007b) showed that despite the influence of nonlinearity, the differences in shape for all bell-shaped positive pulses, such as solitary wave, Lorentz and sinusoidal pulses are negligible and can be parameterized. This

* Corresponding author.
E-mail address: vsriram@iitm.ac.in (V. Sriram).

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Fig. 1. Indian ocean tsunami of 26 December 2004 approaching Koh Jum Island, off the coast of Thailand (Copyright Anders Grawin, 2006).

parameterization is especially important for the estimation of tsunami characteristics at the coast, when the shape of the wave approaching the coast is unknown. Also, the asymmetry of a wave approaching the coast and the steepness of its front leads to significant amplification of wave run-up height and shoreline velocity, which was observed during the catastrophic 2004 Indian Ocean tsunami (Didenkulova et al., 2007a).

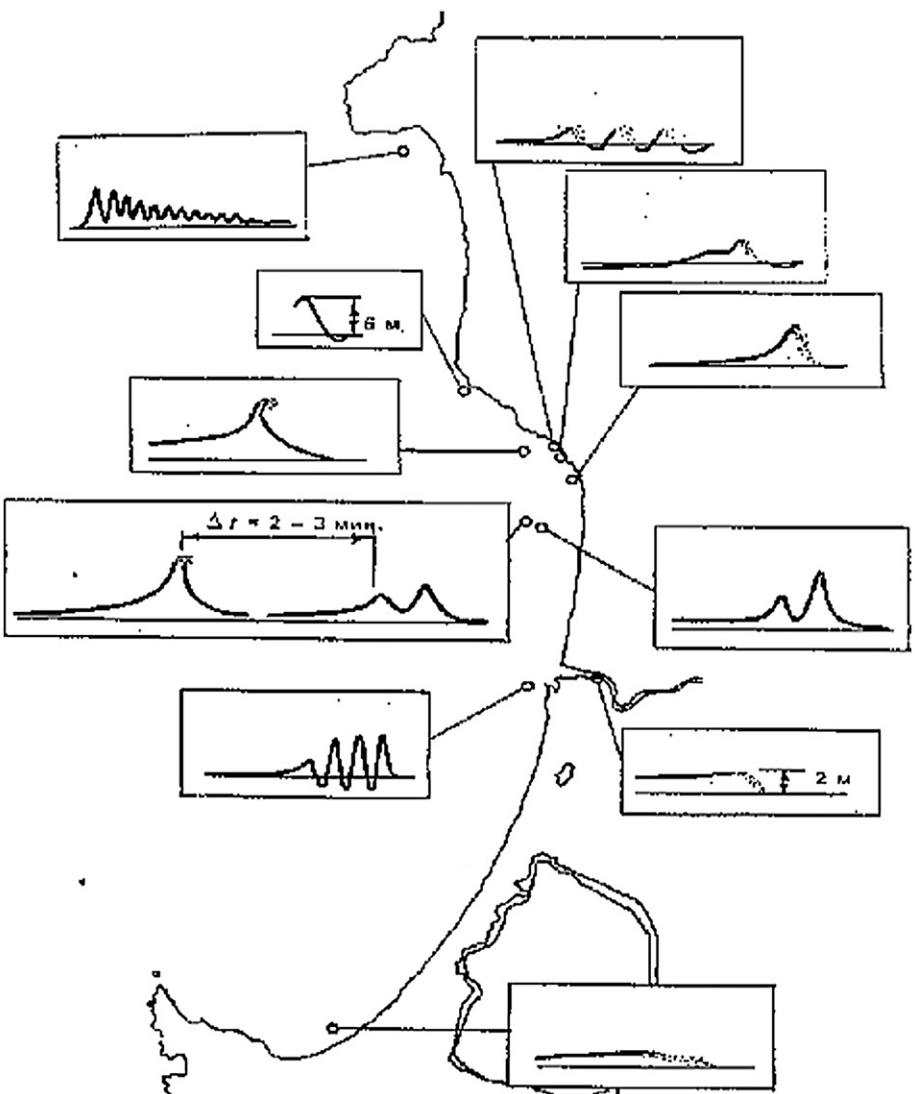

Fig. 2. Tsunami wave shapes at the Japanese coast after the Japan Sea tsunami occurred on 26 May 1983 (Shuto, 1985).

It is also obvious that the bottom and coastal topography has a strong influence on wave run-up characteristics.

It has been demonstrated by Didenkulova et al. (2009) and Didenkulova and Pelinovsky (2010) that some profiles (~$x^{4/3}$ from Tinti et al. (2001) and ~x^4) mathematically correspond to "nonreflecting" wave propagation and can lead to abnormal wave amplification on the beach. The same effects are also observed in narrow bays (Didenkulova and Pelinovsky, 2009, 2011). The developed theory was successfully applied to explain abnormal run-ups during 2009 Samoa tsunami (Didenkulova, 2013) and 2011 Japan tsunami (Kim et al., 2013).

Breaking bores or hydraulic jumps have been widely studied both theoretically and experimentally. A special case, the so-called dam break problem, became a benchmark for numerical simulations (Liu et al., 2008) and is often used for testing the stability of coastal structures (Haehnel and Daly, 2004; Riggs et al., 2013).

As mentioned above, an undular bore is usually observed after long distance propagation, as it was during the Indian Ocean tsunami (see Fig. 1). Hence, it should be studied in facilities which allow long distance wave propagation, such as the Large Wave Flume (Großer Wellenkanal, GWK), Hannover, Germany. As such facilities are just a few, most undular bore studies are numerical (Brühl and Oumeraci, 2014; Grue et al., 2008; Sriram et al., 2007).

It has been demonstrated in the companion paper (Schimmels et al., 2015) that existing testing facilities with piston type wave makers (having a large stroke) can effectively be used to generate long waves, including very long tsunami-like waves. In this paper, we will focus on the propagation and run-up characteristics of these long waves and will consider waves of various shapes including real (scaled down) tsunami records.

The paper is arranged in the following manner. After a brief description of the applied numerical models based on fully nonlinear potential flow theory (FNPT) and Korteweg–de Vries (KdV), the experimental data of wave propagation are compared with the results of numerical simulations. Then, the conditions for undular bore formation are also verified theoretically and finally the same waves/wave trains propagate onshore and climb "hypothetical" beaches of different slopes, confirming that by constructing a mild slope in a large scale experimental facility, one can study the characteristics of tsunami waves of various types.

2. Numerical modeling

2.1. Nonlinear potential flow theory (FNPT)

The fluid is assumed to be incompressible. The flow is irrotational and viscous forces are neglected in this model. This simplifies the flow problem to be defined by Laplace's equation involving a velocity potential $\Phi\,(x,z)$ given by,

$$\nabla^2\Phi = 0. \tag{1}$$

A potential flow in a sub-domain with a wave maker at one end and nonlinear free surface boundary conditions is considered. Prescribed Neumann and Dirichlet boundary conditions are applied on the boundaries. Considering the flume bottom as flat with no flow through it, one can write,

$$\frac{\partial\Phi}{\partial z} = 0 \ \text{ at } \ z = -d \ \text{ on } \ \Gamma_B. \tag{2}$$

Displacement of the wave paddle at the left end can be enforced by,

$$\frac{\partial\Phi}{\partial x} = \dot{x}_p(t) \ \text{ at } \ x = x_p(t) \ \text{ on } \ \Gamma_p, \tag{3}$$

where $x_p(t)$ is the time history of wave paddle displacement at $x = 0$. Thus, measured or theoretical wave paddle displacements can be directly provided into the numerical model. The nonlinear dynamic free-surface condition in Lagrangian system is given by,

$$\frac{dx}{dt} = \frac{\partial\Phi}{\partial x}, \tag{4}$$

$$\frac{dz}{dt} = \frac{\partial\Phi}{\partial z}, \tag{5}$$

$$\frac{d\Phi}{dt} = -P - gz + \frac{1}{2}\nabla\Phi\nabla\Phi, \tag{6}$$

where $P = 0$ and $z = \eta$ on the free surface.

Fig. 3 shows the computational domain with the above specified boundary conditions. The solution for the above boundary value problem (BVP) is sought using a finite element scheme. Formulating the governing Laplace equation constrained with the associated boundary conditions Eqs. (1)–(6) leads to the following finite element system of equations:

$$\int_\Omega \nabla N_i \sum_{j=1}^{m} \phi_j \nabla N_j d\Omega|_{j,i\notin\Gamma_s} = -\int_{\Gamma_p} N_i \dot{x}_p(t) d\Gamma - \int_\Omega \nabla N_i \sum_{j=1}^{m} \phi_j \nabla N_j d\Omega|_{j\in\Gamma_s, i\notin\Gamma_s}, \tag{7}$$

where m is the total number of nodes in the domain, and the potential inside an element $\Phi(x,z)$ can be expressed in terms of its nodal potentials, ϕ_j, as

$$\Phi(x,z) = \sum_{j=1}^{n} \phi_j N_j(x,z). \tag{8}$$

Herein, N_j is the shape function and n is the number of nodes in an element. The above formulation does not have any singularity effect at the intersection point between the free surface and the boundaries, which makes it beneficial compared to other methods like BEM. A linear triangular element with structured mesh is used. For further details about the numerical model the reader is referred to Sriram et al. (2010). Here and after this model will be referenced as FNPT-FEM.

2.2. Korteweg–de Vries (KdV) equation

Complimentary to the nonlinear potential flow model, we also use a shallow water model based on the Korteweg–de Vries (KdV) equation (see Sergeeva et al., 2011 for details):

$$\sqrt{gd}\frac{\partial\eta}{\partial x} + \frac{3\eta}{2d}\frac{\partial\eta}{\partial s} + \frac{d}{6g}\frac{\partial^3\eta}{\partial s^3} = 0, \qquad s = \frac{x - x_0}{\sqrt{gd}} - t. \tag{9}$$

Eq. (9) is solved numerically in a periodic domain for s with a period T using a pseudo-spectral method for time derivation and a Crank–Nicolson scheme for space discretization, see details in Fornberg (1996).

Two integrals of Eq. (9) are conserved:

$$\int_0^T \eta(x,s)ds = const, \qquad \int_0^T \eta^2(x,s)ds = const \ . \tag{10}$$

The solution of Eq. (9) is calculated in the discrete Fourier space for spectral amplitudes and defined through the Fourier transform $C(\omega,x) = \mathrm{FFT}\{\eta\,(s,x)\}$, and the nonlinear part in Eq. (9) is calculated

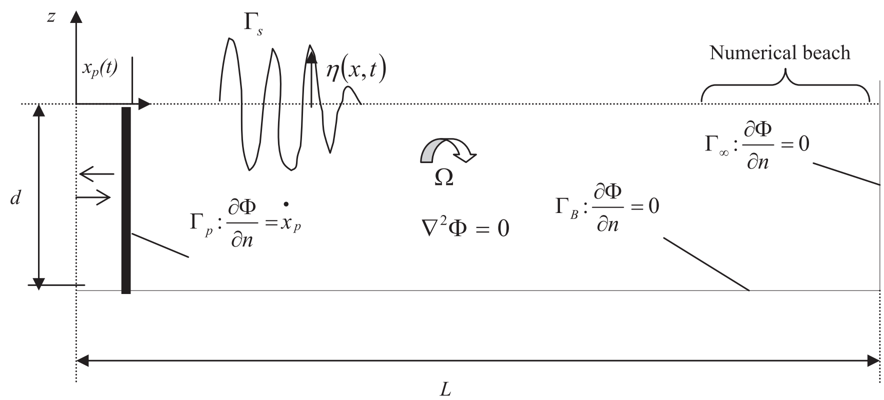

Fig. 3. Computational domain with specified boundary conditions for representing GWK tank.

as $C_{non}(\omega,x) = \mathrm{FFT}\{\eta^2(s,x)/2\}$, where frequency $\omega = 2\pi/T$

$$C(\omega, x+\Delta x) = C(\omega, x-\Delta x)\frac{1+i\beta\omega^3\Delta x}{1-i\beta\omega^3\Delta x} - \frac{3}{2d\sqrt{gd}}C_{non}(\omega,x)\frac{i\omega\Delta x}{1-i\beta\omega^3\Delta x}, \tag{11}$$

where

$$\beta = \frac{d^{1/2}}{6g^{3/2}}. \tag{12}$$

3. Experimental test cases

The large scale experimental study was carried out in the Large Wave Flume (Großer Wellenkanal, GWK) of Forschungszentrum Küste (FZK). A detailed description of the experimental setup and the generation methodology has been given in Schimmels et al. (2015) and will not be repeated here. However, for the completeness of the study, the tested cases are reported in Table 1. For the generation of the waves, Schimmels et al. (2015) used a generic approach by combining several solitons ($sech^2$-waves) as suggested by Chan and Liu (2012):

$$\eta(t) = \sum_{i=1}^{N} H_i sech^2[\omega_i(t-(t_0+t_i))], \tag{13}$$

where N is the total number of solitons; H_i is the wave height of the soliton i; the cyclic frequency of the soliton i is defined through its period, T_i, as $\omega_i = 2\pi/T_i$; t_i is the time shift of soliton i and t_0 is the time shift of the whole wave.

Additionally, we also use another method based on the Self-Correcting Method (SCM) as described in Fernández et al. (2014). Using this method we reproduce downscaled sea level oscillations, measured in Pago Pago harbor during Samoa 2009 tsunami.

4. Long wave propagation

In this section, the results of the previous test cases are reported and discussed along with the numerical modeling based on FNPT-FEM. Since, the generated waves are long, in addition to FNPT-FEM simulations, the Korteweg–de Vries model is also used to compare it with the experimental data. This will give us the range of validity of used numerical models.

The test cases are classified into five different categories based on their shape, namely, regular waves, solitary waves of positive polarity, waves of variable polarity with the leading trough (so-called N-waves with 2 and 3 $sech^2$ profiles) and the down-scaled tsunami time-series.

5. Comparison between experiments and numerical simulations

The length of the tank for the FNPT-FEM simulations was 300 m, with a numerical beach at the far end. The numbers of nodes used in the simulation are 800 and 16 in the x and y directions, respectively with a time step of 0.05 s. This has been kept constant throughout this paper, as it produced acceptable results. The details about the numerical convergence of this model can be referred in Sriram et al. (2006, 2010). The simulations were carried out on a Dell Laptop with 1.28 GB RAM

Table 1
Tested cases.

Test No	H1 (m)	H2 (m)	H3 (m)	T1 (s)	T2 (s)	T3 (s)	t1 (s)	t2 (s)	t3 (s)	Type
1/1	0.06	–	–	100	–	–	–	–	–	Regular wave
1/2	0.08	–	–	60	–	–	–	–	–	Regular wave
2/1	0.04	–	–	30	–	–	–	–	–	1 $sech^2$ profile (elongated soliton)
2/2	0.06	–	–	30	–	–	–	–	–	1 $sech^2$ profile (elongated soliton)
2/3	0.12	–	–	30	–	–	–	–	–	1 $sech^2$ profile (elongated soliton)
2/4	0.11	–	–	30	–	–	–	–	–	1 $sech^2$ Profile (elongated soliton)
2/5	0.1	–	–	30	–	–	–	–	–	1 $sech^2$ Profile (elongated soliton)
2/6	0.05	–	–	30	–	–	–	–	–	1 $sech^2$ Profile (elongated soliton)
3/1	−0.02	0.04	–	100	100	–	90	135	–	2 $sech^2$ profile (Asymmetry)
3/2	−0.04	0.04	–	100	100	–	90	135	–	2 $sech^2$ profile (symmetry)
3/3	−0.05	0.13	–	21	21	–	20	30	–	2 $sech^2$ profile (N-wave)
3/4	−0.05	0.07	–	21	21	–	20	30	–	2 $sech^2$ profile (N-wave)
4/1	−0.05	0.09	0.13	21	21	18	20	30	35	3 $sech^2$ profile (N-wave)
4/2	−0.05	0.05	0.07	21	21	18	20	30	35	3 $sech^2$ profile (N-wave)
5/1	−0.02	0.028	0.04	150	137	41	83.4	111.6	133.8	Iwate profile
5/2	−0.03	0.04	0.01	126	126	63	60	101	140	Mercator profile
5/3	0.12			120						Self-correcting method; Pago Pago, (Fernández et al., 2014)

and it took less than 10 min for each of the reported cases. It should be noted that in the experiments there was a slope of 1:6 at the far end; hence the wave reflection in the experimental and numerical data are different. The main focus of this paper is on propagation of long waves, and serious consideration of numerical absorption can divert this focus. The wave time history from the FNPT-FEM model at 50 m was used to initialize the KdV simulations. We integrate the KdV equation in space for a distance of 300 m with $\Delta x = 0.25$ m. The conservation of the integrals (Eq. (10)) was controlled with fine accuracy.

5.1. Elongated solitary waves

In this section, long solitary waves of positive polarity (elongated solitons) are reported. Experimental investigations on solitons (rigorous solutions of KdV equation) were carried out by many researchers partially due to easiness of their generation in the laboratory (for example, recent papers by Chen and Yeh (2014), Chen et al. (2014), Seiffert et al. (2014), Hayatdavoodi et al.(2014), Lo and Liu, (2014), Lin et al. (2014), Romano et al. (2014), Mo et al. (2013), Wang et al. (2013), Lo et al. (2013), Quiroga and Cheung (2013)). Nevertheless, as quoted in the Introduction (see paper by Madsen et al., 2008 for justification), the period of a tsunami in nature is much longer than the one of KdV solitons. Therefore, in this study we generate elongated solitary waves, which have the same sech^2 shape as classical solitons, but much longer periods. In this section, we consider solitary waves with a period of about 30 s (see Schimmels et al., 2015). The experimental data for test cases 2/1, 2/2, 2/3 (the wave heights are 0.04 m, 0.06 m and 0.12 m, respectively) together with the corresponding numerical simulations obtained in the framework of KdV and FNPT-FEM models are shown in Fig. 4. Both the FNPT-FEM and KdV models are in a good agreement with the experimental measurements. Further, one can clearly see up to 60 m from the wave paddle, all the profiles are stable, thus there is no instability in generation of these long waves using the present methodologies. The second long period wave that is present in all the experiments corresponds to the reflection from the slope. Further, one can see that the wave deformation and disintegration (formation of the undular bore) after propagating long distances is captured by both models accurately (Fig. 4). Thus, the large wave flume allows us in studying both non-breaking waves and undular bores.

5.2. Waves of variable polarity with a leading trough (N-waves)

It has been noticed and proved by Tadepalli and Synolakis (1994) that a wave of variable polarity with a leading trough (so-called *N*-wave) leads to a larger wave amplification and higher run-up on a beach. Later this effect has been interpreted by Didenkulova et al. (2007a, 2007b) as the influence of wave front steepness.

In this study we also consider the propagation of two types of *N*-waves, symmetric and asymmetric. The symmetric *N*-wave consists of two symmetric sech^2 profiles of opposite polarity, whereas, the asymmetric *N*-wave has a twice larger crest compared to its trough. The experimental measurements at 60 m from the wave paddle for these two *N*-waves are shown in Fig. 5. The figure shows good agreement between the FNPT-FEM and experimental measurements. The duration or the period of the wave is about 120 s. Due to the large wavelength, the wave gauges located close to the end of the flume, also register wave reflection from the beach and we do not reproduce their records here. The KdV simulations are also not reproduced here, as the results are identical to FNPT-FEM. Further, the evolution of *N*-waves is studied with respect to their period and height. First, two sech^2 profiles with different wave heights and the same wave period have been tested (test cases 3/3 and 3/4). The wave records at 60 m and 225 m are shown in Fig. 6 and compared with the FNPT-FEM and KdV model results. At longer distance, one can clearly see the formation of an undular bore as a result of the disintegration of the generated waves into a series of solitons. There is a small phase difference between the experiments and the numerical models for the disintegrated waves for the large amplitude waves.

Since, the KdV is based on weak nonlinearity, one can see from Fig. 6b, for weak-amplitude waves, both simulations coincide with the measurements. The difference starts to be visible for higher amplitude

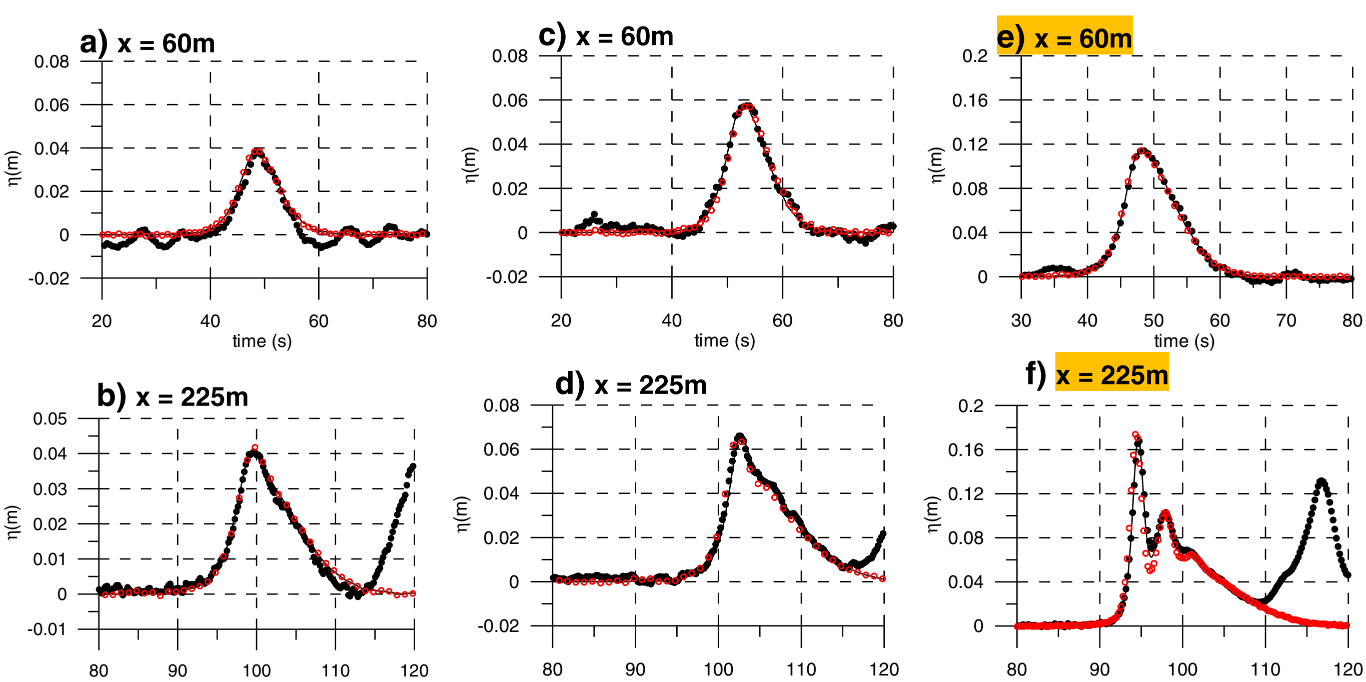

Fig. 4. Comparison between the experiments (filed black circle), FNPT-FEM (solid line) and KdV (red open circle) simulations for the positive wave pulses with heights a) & b) $H = 0.04$ m c) &d) 0.06 m and e) & f) 0.12 m, and wave period $T = 30$ s.

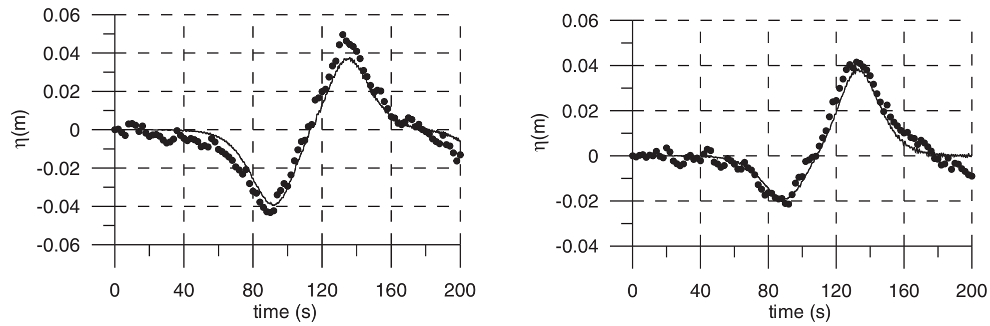

Fig. 5. Symmetric (left) and asymmetric (right) *N*-waves generated at 60 m from the paddle. Line: FNPT-FEM simulation, filled black circles: experiments.

and higher nonlinearity cases (Fig. 6a). For the KdV model these deviations can be related to the influence of nonlinearity. Also, both numerical models do not take into account dissipation and viscous effects, which may also contribute to the observed differences between numerical and experimental records. Particularly, the crest height increases by roughly 1.8 times the initial crest height and the trough height reduces. This shows that there is an energy transfer from trough to crest.

We also consider more complicated composite *N*-waves of three sech^2 profiles (with one being the trough) as it has been done in Chan and Liu (2012) for the best representation of tsunami time-series. The frequency and wave heights of the composite solitons are given in Table 1. Again, two different cases are being analyzed (4/1 and 4/2) with high and small crest heights having the same trough height. Fig. 7 shows the good agreement between experimental, KdV and FNPT-FEM results. As in the previous cases, one can notice the disintegration of waves into a series of multiple waves as the steepness or amplitude increases. The formation of an undular bore as a result of the disintegration or split-up of waves will be discussed in the next section.

6. Wave breaking and undular bore formation

The evolution of long waves in shallow water depends on the wave amplitude (Stoker, 1957; Teles Da Silva and Peregrine, 1990). Large amplitude waves ($\eta_{max}/d \geq 0.5$) propagate following a non-dispersive scenario, getting steeper and steeper during propagation and eventually form a breaking bore. Waves of smaller amplitude are influenced by dispersive effects and evolve into an undular bore. All waves considered in this paper are of small amplitude ($\eta_{max}/d < 0.5$) and, therefore, follow the second scenario, i.e. transformation into a set of solitons of different wavelengths and amplitudes.

However, the formation of a breaking bore in GWK is also possible by changing the bottom profile, e.g. by constructing an underwater step or mild slope. This will be discussed later in this paper. In this section, we will address the formation of undular bores by discussing the distance of actual wave disintegration. The distance where the wave starts to split up can be linked to the wave breaking distance *X* found from the non-dispersive nonlinear shallow water theory (Didenkulova

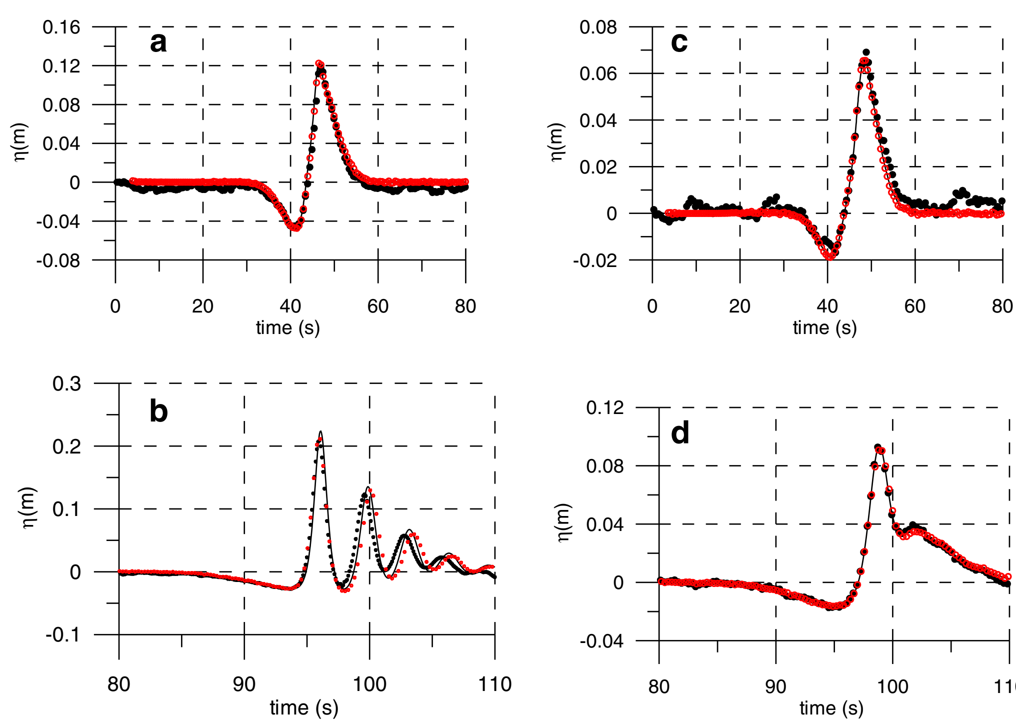

Fig. 6. Propagation of the two sech^2 (.) profile for a long distance. a) Test case 3/3 — Wave history at 60 m from the wave paddle; b) test case 3/3 — Wave history at 225 m from the wave paddle; c) test case 3/4 — Wave history at 60 m from the wave paddle; d) test case 3/4 — Wave history at 225 m from the wave paddle; Line: FNPT-FEM, filed black circles: Experiments; open red circles: KdV simulation.

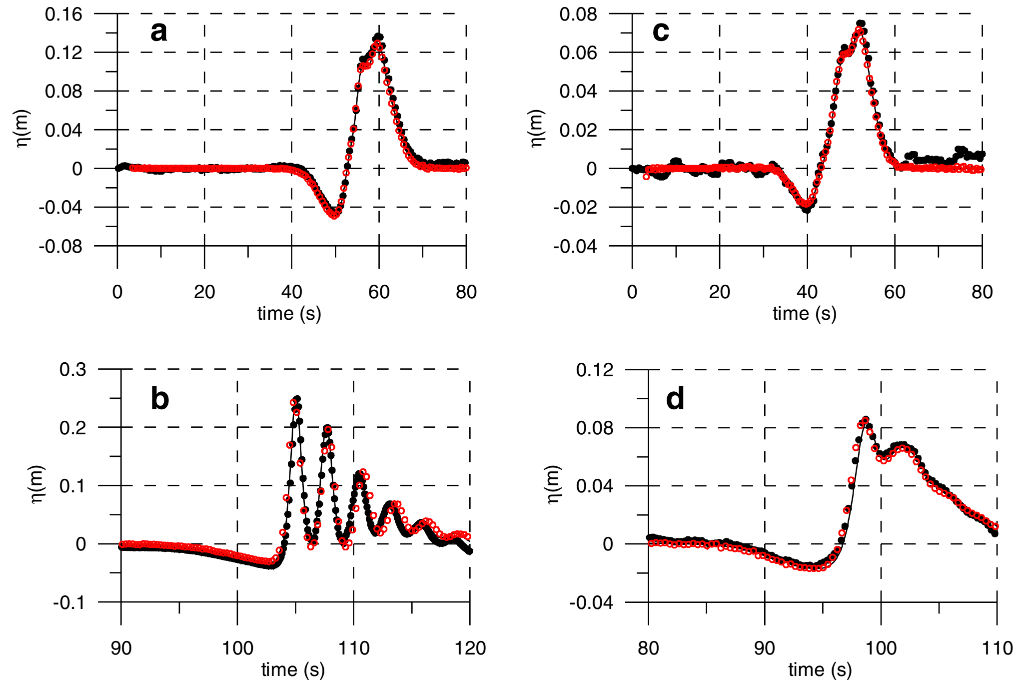

Fig. 7. Propagation of the three sech[2] (.) profile for long distance. a) Test case 4/1 — Wave history at 60 m from the wave paddle; b) test case 4/1 — Wave history at 225 m from the wave paddle c) test case 4/2 — Wave history at 60 m from the wave paddle d) test case 4/2 — Wave history at 225 m from the wave paddle; Line: FNPT-FEM, circle: KdV solution; closed circle: Experiments.

et al., 2006; Zahibo et al., 2008)

$$X = \frac{\sqrt{gd}}{\max(-dV/dt)}, \tag{14}$$

$$V = 3\sqrt{g(d+\eta)} - 2\sqrt{gd}. \tag{15}$$

Eq. (14) will give the propagation distance, before it forms a vertical front and disintegrate. Estimates of the split up distance based on Eq. (14) using the time-series at 50 m are shown in Table 2 (fourth column).

Further, the split up of an initially long wave into solitons is well-described by the KdV equation. For a soliton-like initial wave shape (sech^2) of height H the number of formed solitons and their amplitudes H_n can be found as (Pelinovsky, 1996)

$$H_n = \frac{H}{4Ur}\left[\sqrt{1+8Ur} - (2n-1)\right]^2, \quad n = 1, 2, \ldots N, \tag{16}$$

$$N = \left\lfloor \frac{\sqrt{1+8Ur}+1}{2} \right\rfloor,$$

where Ur is the Ursell number. It can be seen from Eq. (16) that the amplitudes of formed solitons cannot exceed twice the amplitude of the initial wave. For example, in our test case 2/3, for a wave with $H = 0.12$ m and T = 30 s, the largest soliton has an amplitude of 0.19 m. This is in agreement with the numerical simulations at 270 m from the wave paddle, where the first wave height is about 0.185 m.

In order to investigate in detail the split up of the considered waves and due to the lack of data at many experimental locations, the results from FNPT-FEM and KdV simulations at various locations (spacing of 20 m) and the water surface elevation time series are used to analyze the results. This is plotted in Fig. 8 for elongated solitons. These plots show that the splitting of waves occurs at around 150 m for the wave of 0.12 m height, at 250 m for the wave of 0.06 m height and no splitting was noticed up to 300 m for the case of 0.04 m wave height, which is also in a good agreement with the estimates based on Eq. (14), see Table 2. The surprisingly good agreement for split-up distances for all three models suggests that the breaking distance found from Eq. (14) based on nonlinear shallow water theory is a robust characteristic and can also be used as an estimation of split-up distances for long dispersive waves.

Sometimes wave splitting occurs due to the instability in wave generation attributed to an improper paddle motion (Schäffer, 1996). However, for the tested cases, the splitting is in agreement with the theoretical estimates, hence, the generation algorithm, in which the paddle displacement obtained using classical Runge–Kutta method (RK4th order), is fairly reasonable.

In addition to the single pulses shown in Fig. 8, now the evolutions of the N-waves are shown in Fig. 9. Since, the KdV and FNPT simulations gave us identical results, for clarity only FNPT results are shown in Fig. 9. The approximate split up distance shown by the arrows is in agreement with the estimates performed within nonlinear shallow water theory (Table 2).

It can also be seen that the wave trough for N-waves decreases during wave propagation. This effect may be related to the formation of an undular bore, so that the initial N-wave starts to transform into a set of solitons of different wavelengths and amplitudes. After a very long propagation, the trough should vanish and the wave profile will be represented by a set of solitons. This is exactly reproduced using the

Table 2
Split up distances, based on FNPT-FEM and KdV simulations and theoretical estimates.

Test No.	FNPT-FEM split up distance (m)	KdV split up distance (m)	SWT split up distance (m)
2/3	146	146	147
2/2	250	250	251
2/1	340	340	341
3/1	368	368	370
3/2	348	348	346
3/3	100	100	99
3/4	160	160	159
4/1	195	195	193
4/2	115	115	115

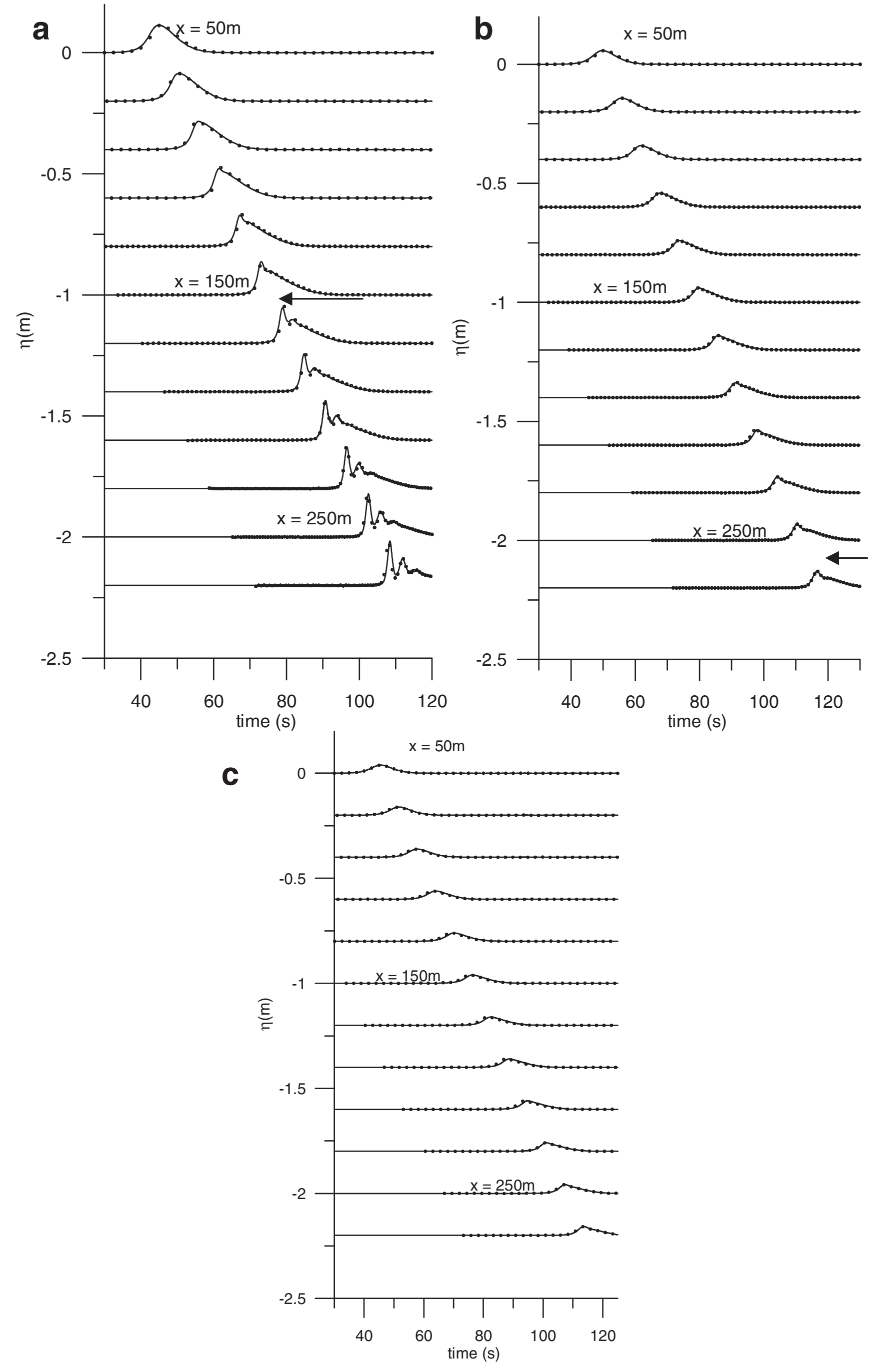

Fig. 8. FNPT-FEM (solid line) and KdV (dots) simulations of wave propagation along the 1 m deep wave flume. a) The wave height of 0.12 m, b) 0.06 m and c) 0.04 m, the period of all waves is 30 s. Arrows represent the numerical split up distances, which are in agreement with the theoretical predictions shown in Table 2.

present numerical model and is in agreement with the experimental measurements.

6.1. Tsunami time-series

In all the cases above, a classical time integration of the target wave has been used to calculate the stroke motion and generate the desired signal. Nevertheless, sometimes one needs a procedure or algorithm to reproduce field measurements at a pre-defined point in the flume, say at 100 m or 200 m. In order to implement this, we have used a self-correcting algorithm introduced by Daemrich et al. (1980) and Daemrich and Gotschenberg (1988). This method has been recently extended and validated by Fernández et al. (2014). We have used the FNPT model to get the required stroke by self-correction or iterations with respect to the target signal. After that, the final stroke has been given to the GWK wave paddle to generate the same. This algorithm has been applied to the time-series recorded by the tide-gauge during Samoa 2009 tsunami in Pago Pago harbor (Didenkulova, 2013; Zhou et al., 2012), scaled by 1:37. The correction steps are shown in Fig. 10. For this case, after the 2nd correction, no more improvements have been found (Fernández et al., 2014). The target location for this case is at 225 m from the wave paddle.

It should be pointed out that for long regular waves there was reflection from the slope at this location (Schimmels et al., 2015). Hence, we have chosen this location in order to show the capability of the self-correction method. This final stroke has been used in GWK. The results

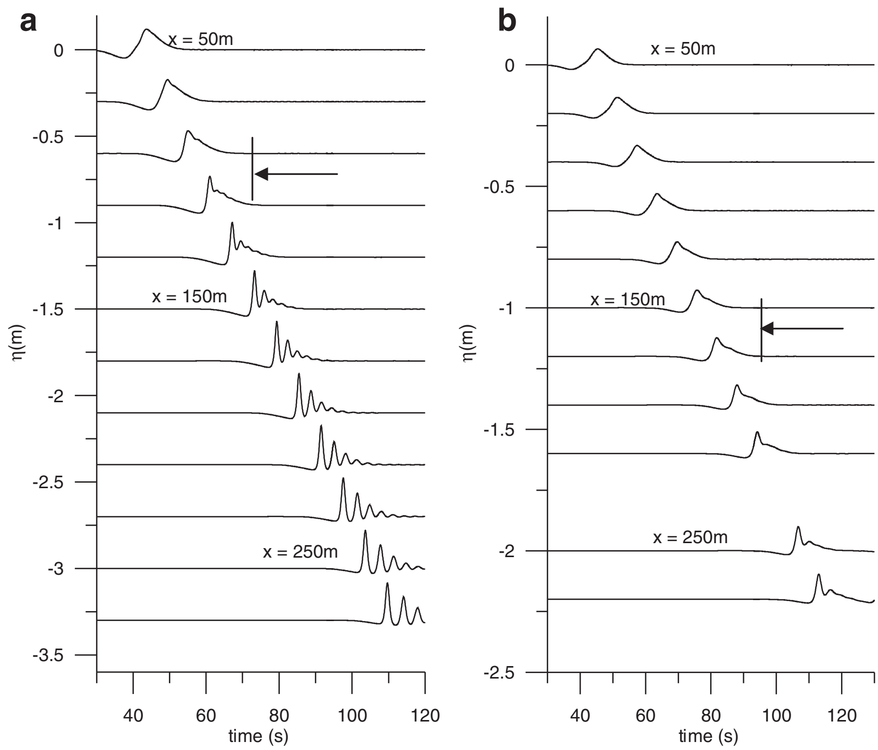

Fig. 9. FNPT-FEM simulation of the wave propagation at different locations, showing the splitting of the wave forms. The arrow represents the numerical splitting locations, which is in agreement with the theoretical predictions shown in Table 2. a) Test case 3/1, b) test case 3/2.

from GWK as well as the numerical simulations are plotted in Fig. 11, showing good agreement with the target profile (field data). The self-correction in FNPT model for the present case took roughly 20 min, thus it is a highly economical and reproducible tool for the large scale experiments. For more details see Fernández et al. (2014).

7. Run-up of long waves in GWK

As it has been mentioned before, one of the main questions in tsunami related studies are run-up and its impact on the coast. From this perspective, it is extremely important to make sure that the generated

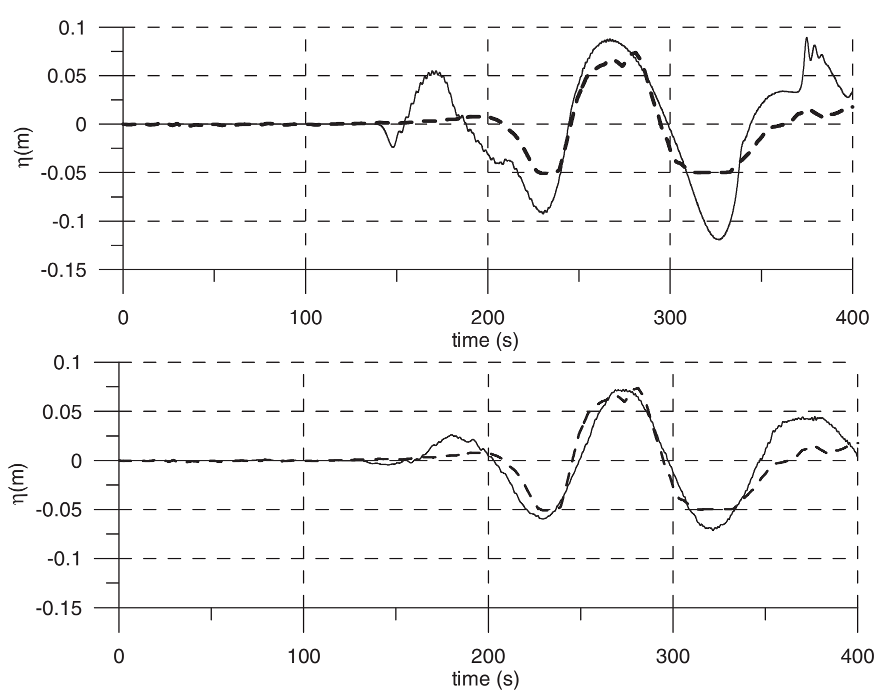

Fig. 10. Self correction in FNPT for Pago Pago tsunami record at target location. Top: wave elevation from the 1st correction. Bottom plot: Wave elevation from the 2nd correction (at 225 m). Solid line: FNPT-FEM, dashed line — target tide gauge record (Refer Fernández et al., 2014).

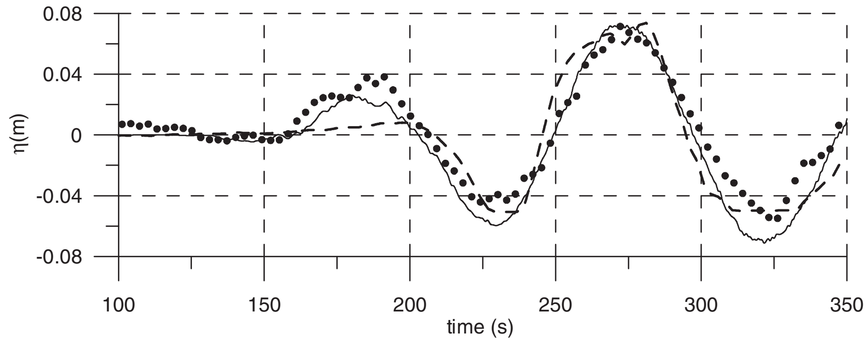

Fig. 11. Comparison of the GWK, FNPT results at prescribed location with target Pago Pago tsunami-record. Solid line: FNPT-FEM, dashed line — target tide gauge record, dotted — Experiments.

waves can also be used for this kind of studies. Based on the current data, one can easily question the height of the generated waves, which was rather small, and might not lead to sufficient run-up.

The natural slope in GWK is quite steep 1:6. Considering our water depth of 1 m the length of the slope is just 6 m, which is too short compared to the length of the considered waves, the shortest of which has a wavelength of 60 m. It is clear that such long waves do not really feel the slope and cannot amplify significantly while climbing the beach. However, the typical beach slopes are usually much milder. For example, the beach slope in Pago Pago harbor was 1:30. Therefore, in order to reproduce a realistic run-up, one should decrease the slope angle (α).

7.1. Run-up of Pago Pago tsunami

In this section, the run-up of the tsunami time-series in Pago Pago described in the previous section for a hypothetical slope has been discussed. Oscillations of the moving shoreline (wave run-up) are calculated using asymptotic solutions of the nonlinear shallow water equations following a two-step procedure described in Didenkulova (2009). In this study, we consider two types of beach slope 1:12 and 1:18. Slopes milder than 1:18 lead to wave breaking during their run-up, which cannot be handled analytically. However, in order to model different breaking characteristics, one can use numerical model, such as a hybrid method recently developed in Sriram et al. (2014). In the first stage, the solution of the linear shallow water equations is found using a Fourier integral:

$$R(t) = \sqrt{\frac{4\pi L}{\sqrt{gd}}}\int_{-\infty}^{\infty} \sqrt{\omega} A(\omega) \exp\left[i\left(\omega t + \frac{\pi}{4} sign(\omega)\right)\right] d\omega, \tag{17}$$

$$U(t) = \frac{1}{\tan(\alpha)} \frac{dR}{dt},$$

where $A(\omega)$ are complex amplitudes found from the incident wave measurements at the toe of the slope η_{sl}

$$\eta_{sl}(t) = \int_{-\infty}^{\infty} A(\omega) \exp(i\omega t) d\omega, \tag{18}$$

L is the length of the slope, R and U are the water displacement and velocity at the point of zero water depth.

In the second stage, the real run-up oscillation r and shoreline velocity u using the following relations are evaluated.

$$r(t) = R\left(t + \frac{u}{\tan(\alpha)g}\right) - \frac{u^2}{2g}, \qquad u(t) = U\left(t + \frac{u}{\tan(\alpha)g}\right), \tag{19}$$

where α is the beach slope. It should be noted that this procedure is limited to the case of non-breaking wave propagation and run-up. Hence, only a limited range of possible beach angles could be considered. This condition is defined by the parameter Br, which must not be larger than 1 ($Br \le 1$)

$$Br = \frac{1}{g \tan^2(\alpha)} \max\left[\frac{d^2R}{dt^2}\right]. \tag{20}$$

The run-up heights and inundation lengths for the Pago Pago tsunami time-series on a plane beach are given in Fig. 12 for slopes of 1:12 (red dashed line) and 1:18 (black solid line). Maximum run-up heights on these two beaches reach 0.20 m and 0.24 m, which corresponds to an amplification of 2.7 and 3.2 times with respect to the maximum wave amplitude at the toe of the slope (~0.07 m). The maximum inundation length at the 1:18 slope is larger than 4 m. If one constructs a much milder slope than this (and the size of GWK facility allows us to do so in a wide range of slopes, up to 1:200), the waves will experience breaking and at certain conditions will form a bore propagating to the coast. In this case we will be able to study the third type of tsunami waves (breaking bore or hydraulic jump) as described in the introduction. Another way to form a breaking bore would be to build up an underwater step and then let the bore run-up the beach. Thus, large scale facilities would provide a wide range of opportunities to study run-up of "real" tsunami waves of various types.

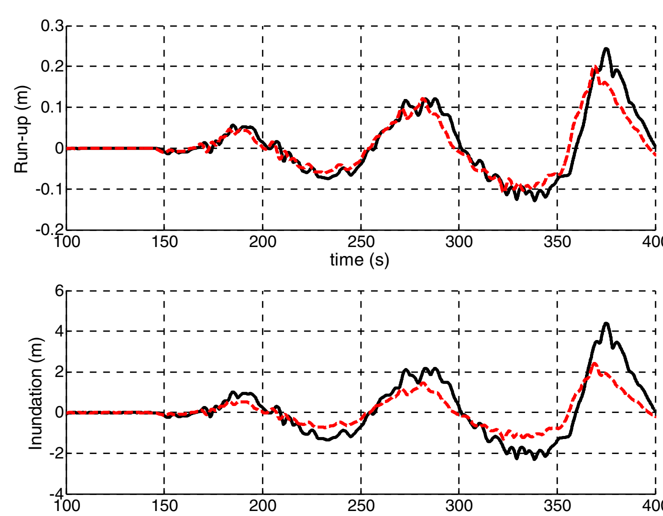

Fig. 12. Run-up height (top) and inundation length (bottom) of Pago Pago tsunami time-series on a plane beach with a slope 1:12 (red dashed line) and 1:18 (black solid line).

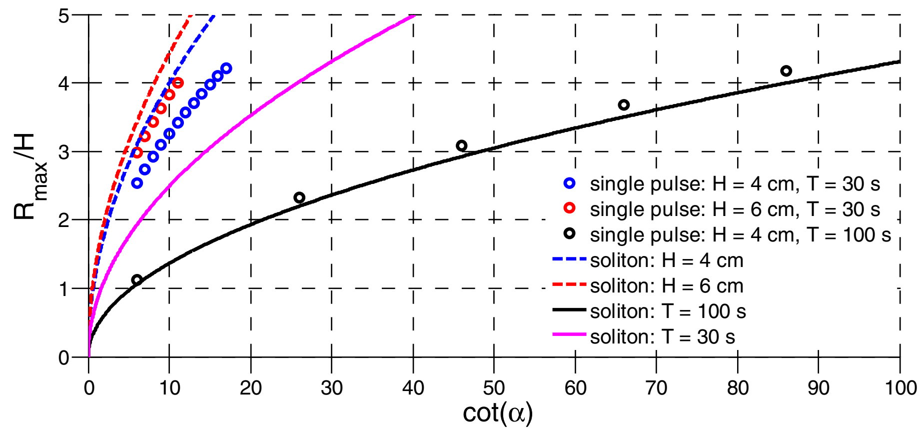

Fig. 13. Run-up height of elongated solitary waves with H = 4 cm and T = 30 s (blue circles), H = 6 cm and T = 30 s (red circles), H = 4 cm and T = 100 s (black circles) on beaches with different slopes; dashed lines correspond to run-up of solitons by Eq. (21) for H = 6 cm (red) and 4 cm (blue); solid lines correspond to run-up of solitons by Eq. (22) for T = 30 s (magenta) 100 s (black).

7.2. Run-up of single pulses

In this section, run-up of the elongated solitons (i.e. single pulses) over different slopes has been investigated. Run-up in terms of the ratio of maximum run-up height R_{max} and the initial wave height, H is shown in Fig. 13 for waves of two different amplitudes (4 cm and 6 cm) and two different periods (30 s and 100 s). As in the previous section, the hypothetical wave run-up in Fig. 13 is considered only in the range of non-breaking waves. Since the run-up height of non-breaking waves is inversely proportional to T^2, the shorter waves are amplified stronger but also break sooner. This is manifested in a narrower range of considered beach slopes for waves of shorter periods, as can be seen in Fig. 13. As it has already been mentioned in the previous section, this feature and the size of GWK facility together give us an opportunity to study such long non-breaking waves and long breaking bore waves in a wide range of parameters.

For comparison, Synolakis (1987) solitary wave formulae have been used, which describes run-up of solitons with amplitudes of 4 cm and 6 cm on a plane beach:

$$\frac{R_{max}}{H} = 2.8312\sqrt{\cot(\alpha)}\left(\frac{H}{d}\right)^{1/4}. \tag{21}$$

The run-up heights of solitons with amplitudes 4 cm and 6 cm calculated by Eq. (21) are plotted in Fig. 13 by dashed lines. It can be seen that the Synolakis' formula for solitons overestimates the actual run-up heights of the considered pulses. As wave height and period for solitons are interconnected, we can select the parameter for comparison. In Eq. (21) this parameter was the wave height. If we try to compare solitons with the single pulses by the period, Eq. (21) can be re-written as

$$\frac{R_{max}}{\eta_{max}} = 2.8312\sqrt{\cot(\alpha)}\left(\frac{1}{gd}\left(\frac{4\pi d}{\sqrt{3}T}\right)^2\right)^{1/4}. \tag{22}$$

The run-up of solitons with periods of 30 s and 100 s calculated by Eq. (22) are plotted in Fig. 13 by solid lines. It can be seen that if one use a period as the major parameter, the comparison with the run-up heights of the studied long period waves is better, but still far from the ideal and leads to an underestimation of the actual heights. This demonstrates again that a one-parameter solitary wave cannot be a good model for such a complex multi-parameter phenomenon like tsunami and represents just a limiting case. For a reasonable description of the tsunami phenomenon one needs at least a two-parameter wave (one parameter is the wave height and another is the period). Indeed, as it has been shown in (Didenkulova et al., 2007a) for nonlinear deformed waves and in Tadepalli and Synolakis (1994) for *N*-waves, in some cases even a two-parameter description is not enough and one needs to have a third parameter, which could be a wave steepness, for instance. Thus, here we want to underline that in large scale facilities, due to its size and the developed wave generation techniques, all three parameters for wave description and generation can be accounted for. The third parameter – asymmetry (steepness) of the wave – can be accounted for by the natural deformation during long distance propagation in GWK.

8. Conclusions

The potential of large scale facilities and in particular the 300 meter long Large Wave Flume (GWK) for tsunami wave propagation and run-up has been examined. Three types of long waves: non-breaking surging waves, breaking bore (or hydraulic jump) and undular bore (or split-up of waves) have been considered in the feasibility analysis. The generation of non-breaking waves has been demonstrated by the example of a downscaled tsunami time-series and small-amplitude elongated solitary waves. For solitary waves of higher amplitude and N-waves, we could see the formation of an undular bore. Due to the stroke limitations, long waves of higher amplitude are not feasible to be generated. Nevertheless, as the length of the tank is large, we believe that by constructing a mild slope or underwater step, we can also generate a breaking bore. The last consideration was also supported by calculations of wave run-up heights at different hypothetical beach slopes that can be constructed in large flumes. Thus, it has been demonstrated that existing large scale test facilities like GWK can reproduce all three types of tsunami waves and can be used in tsunami research to investigate their run-up and inundation characteristics. The option to conduct experimental studies in large scale facilities has a special importance when numerical simulations are costly, time consuming or not possible.

The obtained experimental measurements have been used for verification of numerical codes based on FNPT-FEM and KdV equations. The simulations showed good agreement with the long wave measurements. The process of undular bore formation is well described by both experimental and numerical data. The number of solitons and their split up distance increase as the wave steepness (the steepness of the wave front) increases. For *N*-waves with a leading trough a decrease in the trough height during wave propagation has been noticed. This effect often occurring in shallow water, suggests that there is transfer of the wave energy, which needs further investigations. A comparison of run-up heights of long period pulses and standard soliton solutions confirms the idea of Madsen et al. (2008) that long waves such as tsunami should not be modeled by solitary waves. The experimental measurements for elongated solitons and N-waves carried out in this study has been given as a supplementary data, which may act as a benchmark test cases for validation of the numerical models or theory.

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